

Calculus II: Complete Notes

Illya Starikov

Missouri S&T

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6 Applications of Integration

6.1 Physical Applications

Mass of a One-Dimensional Object

Suppose a thin bar or wire is represented by a line segment on the interval $a \leq x \leq b$ with a density function ρ (with units of mass per length). The mass of the object is

$$m = \int_a^b \rho(x) dx \quad (1)$$

Work

The work done by a variable force F in moving an object along a line from $x = a$ to $x = b$ in the direction of the force is

$$W = \int_a^b F(x) dx \quad (2)$$

Solving Lifting Problems

1. Draw a y -axis in the vertical direction (parallel to gravity) and choose a convenient origin. Assume the interval $[a, b]$ corresponds to the vertical extent of the fluid.
2. For $a \leq y \leq b$, find the cross-sectional area $A(y)$ of the horizontal slices and the distance $D(y)$ the slices must be lifted.
3. The work required to lift the water is

$$W = \int_a^b \rho g A(y) D(y) dy \quad (3)$$

Solving Force/Pressure Problems

1. Draw a y -axis on the face of the dam in the vertical direction and choose a convenient origin (often taken to be the base of the dam).

2. Find the width function $w(y)$ for each value of y on the face of the dam.
3. If the base of the dam is at $y = 0$ and the top of the dam is at $y = a$, then the total force on the dam is

$$F = \int \rho g(a - y)w(y) dy \quad (4)$$

7 Logarithmic, Exponential, and Inverse Trigonometric Functions

7.1 Inverse Function

Derivative of the Inverse Function

Let f be differentiable and have an inverse on an interval I . If x_0 is a point of I at which $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ and

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}, \text{ where } y_0 = f(x_0) \quad (5)$$

7.2 The Natural Logarithmic and Exponential Functions

The Natural Logarithm

The **natural logarithm** of a number $x > 0$, denoted $\ln x$, is defined

$$\ln x = \int_1^x \frac{dt}{t} \quad (6)$$

Properties of the Natural Logarithm

1. The domain and range of $\ln x$ are $(0, \infty)$ and $(-\infty, \infty)$, respectively.
2. $\ln(xy) = \ln x + \ln y, \forall x, y \in \mathbf{R}^+$
3. $\ln(x/y) = \ln x - \ln y, \forall x, y \in \mathbf{R}^+$
4. $\ln x^p = p \ln x, \forall x \in \mathbf{Q}^+$
5. $\frac{d}{dx}(\ln |x|) = \frac{1}{x}, \forall x \in \mathbf{R} \wedge x \neq 0$
6. $\frac{d}{dx}(\ln |u(x)|) = \frac{u'(x)}{u(x)}$
7. $\int \frac{dx}{x} = \ln |x| + C$

The Number e

The number e is the real number that satisfies

$$\ln e = \int_1^e \frac{dt}{t} = 1 \quad (7)$$

The Exponential Function

$\forall x, y \in \mathbf{R}$

$$y = e^x \Leftrightarrow x = \ln y \quad (8)$$

Properties of e

- $e^{x+y} = e^x e^y$
- $e^{x-y} = e^x / e^y$
- $(e^x)^y = e^{xy}, \forall y \in \mathbf{Q}$
- $\ln(e^x) = x, \forall x \in \mathbf{R}$
- $e^{\ln x} = x, \forall x \in \mathbf{R}^+$

Exponential Functions with General Bases

Let $b \in \mathbf{R}^+ \wedge b \neq 1$. $\forall x \in \mathbf{R}^+$,

$$b^x = e^{x \ln b} \quad (9)$$

Derivative and Integral of the Exponential Function

$\forall x \in \mathbf{R}$,

$$\frac{d}{dx}(e^{u(x)}) = e^{u(x)} u'(x) \quad (10)$$

$$\int e^x dx = e^x + C \quad (11)$$

7.3 Logarithmic and Exponential Functions with Other Bases

Logarithmic Function Base b

For any base $b > 0$, with $b \neq 1$, the **logarithmic function base b** , denoted $\log_b x$, is the inverse of the exponential function b^x .

Inverse Relations for Exponential and Logarithmic Functions

For any base $b > 0$, with $b \neq 1$, the following inverse relation holds.

- $b^{\log_b x} = x, \forall x \in \mathbf{R}^+$
- $\log_b b^x = x, \forall x$

Derivative of b^x

If $b > 0 \wedge b \neq 1, \forall x$,

$$\frac{d}{dx}(b^x) = b^x \ln b \quad (12)$$

Indefinite Integral of b^x

For $b > 0 \wedge b \neq 1$,

$$\int b^x dx = \frac{1}{\ln b} b^x + C \quad (13)$$

General Power Rule

$\forall p, x \in \mathbf{R}^+$,

$$\frac{d}{dx}(x^p) = px^{p-1} \quad (14)$$

Furthermore, if u is a positive differentiable function on its domain, then

$$\frac{d}{dx}(u(x)^p) = p(u(x))^{p-1} \times u'(x) \quad (15)$$

Derivative of $\log_b x$

If $b > 1$,

$$\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b} \quad \forall x \in \mathbf{R}^+ \wedge x \neq 0 \quad (16)$$

$$\frac{d}{dx}(\log_b |x|) = \frac{1}{x \ln b} \quad \forall x \in \mathbf{R}^+ \wedge x \neq 0 \quad (17)$$

7.4 Inverse Trigonometric Functions

Derivative of Inverse Trigonometric Functions

$$\begin{aligned}\frac{d}{dx}(\sin^{-1} x) &= \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}(\cos^{-1} x) &= -\frac{1}{\sqrt{1-x^2}}, \{x \in \mathbf{R} \mid -1 < x < 1\} \\ \frac{d}{dx}(\tan^{-1} x) &= \frac{1}{1+x^2} & \frac{d}{dx}(\cot^{-1} x) &= -\frac{1}{1+x^2}, \{x \in \mathbf{R} \mid -\infty < x < \infty\} \\ \frac{d}{dx}(\sec^{-1} x) &= \frac{1}{|x|\sqrt{x^2-1}} & \frac{d}{dx}(\csc^{-1} x) &= -\frac{1}{|x|\sqrt{x^2-1}}, \{x \in \mathbf{R} \mid |x| > 1\}\end{aligned}$$

Integrals Involving Inverse Trigonometric Functions

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C \quad (18)$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C \quad (19)$$

$$\int \frac{dx}{x\sqrt{a^2 + x^2}} = \frac{1}{a} \sec^{-1} \left| \frac{x}{a} \right| + C \quad (20)$$

7.5 L'Hôpital's Rule and Growth Rates of Functions

Indeterminate forms $1^\infty, 0^0, \infty^0$

Assume $\lim_{x \rightarrow \infty} f(x)^{g(x)}$ has the indeterminate form $1^\infty, 0^0$, or ∞^0 .

1. Evaluate $L = \lim_{x \rightarrow \infty} g(x) \ln f(x)$. This limit can be put in the form $0/0$ or ∞/∞ , both of which are handled by l'Hôpital's rule.
2. Then $\lim_{x \rightarrow \infty} f(x)^{g(x)} = e^L$

Growth Rates of Functions (as $x \rightarrow \infty$)

Suppose f and g are functions with $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = \infty$. Then f **grows faster than g** as $x \rightarrow \infty$ if

$$\lim_{x \rightarrow \infty} \frac{g(x)}{f(x)} = 0 \quad \text{or, quantitatively, if} \quad \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \infty \quad (21)$$

The functions f and g have **comparable growth rates** if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = M$$

where $M \in \mathbb{R}^+$.

Ranking Growth Rates as $x \rightarrow \infty$

Let $f \ll g$ mean that g grows faster than f as $f \rightarrow \infty$. With positive real numbers p, q, r, s and $b > 1$,

$$\ln^q x \ll x^p \ll x^p \ln^r x \ll x^{p+s} \ll b^x \ll x^x \quad (22)$$

7.6 Hyperbolic Functions

Hyperbolic Functions

Hyperbolic Cosine

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad (23)$$

Hyperbolic Sine

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad (24)$$

Hyperbolic Tangent

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (25)$$

Hyperbolic Cotangent

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad (26)$$

Hyperbolic Secant

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} \quad (27)$$

Hyperbolic Cosecant

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}} \quad (28)$$

Hyperbolic Identities

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\coth^2 x - 1 = \operatorname{csch}^2 x$$

$$\coth(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$

$$\cosh^2 x = \frac{\cosh 2x + 1}{2}$$

$$\cosh(-x) = \cosh x$$

$$\sinh(-x) = -\sinh x$$

$$\tanh(-x) = -\tanh x$$

$$\sinh 2x = 2 \sinh x \cosh x$$

$$\sinh^2 x = \frac{\cosh 2x - 1}{2}$$

Derivatives and Integral Formulas

1. $\frac{d}{dx}(\cosh x) = \sinh x \Rightarrow \int \sinh x \, dx = \cosh x + C$
2. $\frac{d}{dx}(\sinh x) = \cosh x \Rightarrow \int \cosh x \, dx = \sinh x + C$
3. $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x \Rightarrow \int \operatorname{sech}^2 x \, dx = \tanh x + C$
4. $\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x \Rightarrow \int \operatorname{csch}^2 x \, dx = -\coth x + C$
5. $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x \Rightarrow \int \operatorname{sech} x \tanh x \, dx = -\operatorname{sech} x + C$
6. $\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x \Rightarrow \int \operatorname{csch} x \coth x \, dx = -\operatorname{csch} x + C$

Integrals of Hyperbolic Functions

1. $\int \tanh x \, dx = \ln \cosh x + C$
2. $\int \coth x \, dx = \ln |\sinh x| + C$
3. $\int \operatorname{sech} x \, dx = \tan^{-1} |\sinh x| + C$
4. $\int \operatorname{csch} x \, dx = \ln |\tanh(x/2)| + C$

Inverses of the Hyperbolic Functions Expressed as Logarithms

$$\begin{aligned}\cosh^{-1} x &= \ln(x + \sqrt{x^2 - 1}) \quad (x \geq 1) & \operatorname{sech}^{-1} x &= \cosh^{-1} \frac{1}{x} \quad (0 < x \leq 1) \\ \sinh^{-1} x &= \ln(x + \sqrt{x^2 + 1}) & \operatorname{csch}^{-1} x &= \sinh^{-1} \frac{1}{x} \quad (x \neq 0) \\ \tanh^{-1} x &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \quad (|x| < 1) & \coth^{-1} x &= \tanh^{-1} \frac{1}{x} \quad (|x| > 1)\end{aligned}$$

Derivatives of the Inverse Hyperbolic Functions

$$\begin{aligned}\frac{d}{dx}(\cosh^{-1} x) &= \frac{1}{\sqrt{x^2 - 1}} \quad (x > 1) & \frac{d}{dx}(\sinh^{-1} x) &= \frac{1}{\sqrt{x^2 + 1}} \\ \frac{d}{dx}(\tanh^{-1} x) &= \frac{1}{1 - x^2} \quad (|x| < 1) & \frac{d}{dx}(\coth^{-1} x) &= \frac{1}{1 - x^2} \quad (|x| > 1) \\ \frac{d}{dx}(\operatorname{sech}^{-1} x) &= -\frac{1}{x\sqrt{1 - x^2}} \quad (0 < x < 1) & \frac{d}{dx}(\operatorname{csch}^{-1} x) &= -\frac{1}{|x|\sqrt{1 + x^2}} \quad (x \neq 0)\end{aligned}$$

Integral Formulas

1. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1} \frac{x}{a} + C, \text{ for } x > a$
2. $\int \frac{dx}{x^2 + a^2} = \sinh^{-1} \frac{x}{a} + C, \text{ for all } x$

3. $\int \frac{dx}{a^2-x^2} = \frac{1}{a} \tanh^{-1} \frac{x}{a} + C$, for $|x| < a = \frac{1}{a} \coth^{-1} \frac{x}{a} + C$, for $|x| > a$
4. $\int \frac{dx}{x\sqrt{a^2-x^2}} = -\frac{1}{a} \operatorname{sech}^{-1} \frac{x}{a} + C$, for $0 < x < a$
5. $\int \frac{dx}{x\sqrt{a^2+x^2}} = -\frac{1}{a} \operatorname{csch}^{-1} \frac{|x|}{a} + C$, for $x \neq 0$

8 Integration Techniques

8.1 Basic Approaches

$$\int k \, dx = kx + C \quad (29)$$

$$\int k^p \, dx = \frac{k^{p+1}}{p+1} + C, p \in \mathbf{R} \wedge \neq -1 \quad (30)$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax + C \quad (31)$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax + C \quad (32)$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax + C \quad (33)$$

$$\int \csc^2 ax \, dx = -\frac{1}{a} \cot ax + C \quad (34)$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax + C \quad (35)$$

$$\int \csc ax \cot ax \, dx = -\frac{1}{a} \csc ax \cot ax + C \quad (36)$$

$$\int e^{ax} \, dx = \frac{1}{a} e^{ax} + C \quad (37)$$

$$\int \frac{dx}{x} = \ln |x| + C \quad (38)$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C \quad (39)$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C \quad (40)$$

$$\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \left| \frac{x}{a} \right| + C \quad (41)$$

8.2 Integration By Parts

Suppose that u and v are differentiable functions. Then

$$\int u \, dv = uv - \int v \, du \quad (42)$$

Integration by Parts for Definite Integrals

Let u and v be differentiable. Then

$$\int_a^b u(x)v'(x) \, dx = u(x)v(x) \Big|_a^b - \int_a^b v(x)u'(x) \, dx \quad (43)$$

Integral of $\ln x$

$$\int \ln x \, dx = x \ln x - x + C \quad (44)$$

8.3 Trigonometric Integrals

$\int \sin^m x \cos^n x \, dx$ **Strategy**

m is odd, n real Split off $\sin x$, rewrite the resulting even power of $\sin x$ in terms of $\cos x$, and then use $u = \cos x$

n odd, m real Split off $\cos x$, rewrite the resulting even power of $\cos x$ in terms of $\sin x$, and then use $u = \sin x$.

m and n both even, nonnegative Use half-angle identities to transform the integrand into a polynomial in $\cos 2x$, and apply the preceding strategies once again to powers of $\cos 2x$ greater than 1.

Reduction Formulas

Assume n is a positive integer.

$$\int \sin^n x \, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx \quad (45)$$

$$\int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx \quad (46)$$

$$\int \tan^n x \, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx, n \neq 1 \quad (47)$$

$$\int \sec^n x \, dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx, n \neq 1 \quad (48)$$

Integrals of $\tan x$, $\cot x$, $\sec x$, and $\csc x$

$$\int \tan x \, dx = -\ln |\cos x| + C = \ln |\sec x| + C \quad (49)$$

$$\int \cot x \, dx = \ln |\sin x| + C \quad (50)$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C \quad (51)$$

$$\int \csc x \, dx = -\ln |\csc x + \cot x| + C \quad (52)$$

$\int \tan^m x \sec^n x \, dx$ **Strategy**

n **even** Split off $\sec^2 x$, rewrite the remaining even power of $\sec x$ in terms of $\tan x$, and use $u = \tan x$.

m **odd** Split off $\sec x \tan x$, rewrite the remaining even power of $\tan x$ in terms of $\sec x$, and use $u = \sec x$.

m **even and** n **odd** Rewrite the even power of $\tan x$ in terms of $\sec x$ to produce a polynomial in $\sec x$; apply reduction formula 4 to each term.

8.4 Trigonometric Substitutions

The Integral Contains...	Corresponding Substitution	Useful Identity
$a^2 - x^2$	$x = a \sin \theta, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \forall x \leq a$	$a^2 - a^2 \sin^2 \theta = a^2 \cos^2 \theta$
$a^2 + x^2$	$x = a \tan \theta, -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$a^2 + a^2 \tan^2 \theta = a^2 \sec^2 \theta$
$x^2 - a^2$	$x = a \sec \theta,$	$a^2 \sec^2 \theta - a^2 = a^2 \tan^2 \theta$
	$\begin{cases} 0 \leq \theta < \frac{\pi}{2}, \forall x \geq a \\ \frac{\pi}{2} < \theta \leq \pi, \forall x \leq -a \end{cases}$	

8.5 Partial Fractions

Partial Fractions with Simple Linear Factors

Suppose $f(x) = p(x) > q(x)$, where p and q are polynomials with no common factors and with the degree of p less than the degree of q . Assume that q is the product of simple linear factors. The partial fraction decomposition is obtained as follows.

Step 1. Factor the denominator q in the form $(x - r_1)(x - r_2) \cdots (x - r_n)$, where $r_1, \dots, r_n$ are real numbers.

Step 2. Partial fraction decomposition Form the partial fraction decomposition by writing

$$\frac{p(x)}{q(x)} = \frac{A_1}{x - r_1} + \frac{A_2}{x - r_2} + \cdots + \frac{A_n}{x - r_n} \quad (53)$$

Step 3. Clear denominators Multiply both sides of the equation in Step 2 by $q(x) = (x - r_1)(x - r_2) \cdots (x - r_n)$, which produces conditions for $A_1, \dots, A_n$.

Step 4. Solve for coefficients Equate like powers of x in Step 3 to solve for the undetermined coefficients $A_1, \dots, A_n$.

Partial Fractions For Repeated Linear Factors

Suppose the repeated linear factor $(x - r)^m$ appears in the denominator of a proper rational function in reduced form. The partial fraction decomposition has a partial fraction for each power of $(x - r)$ up to and including the m th power; that is, the partial fraction decomposition contains the sum

$$\frac{A_1}{(x - r)} + \frac{A_2}{(x - r)^2} + \frac{A_3}{(x - r)^3} + \cdots + \frac{A_m}{(x - r)^m} \quad (54)$$

where $A_1, \dots, A_m$ are constants to be determined.

Partial Fractions with Simple Irreducible Quadratic Factors

Suppose a simple irreducible factor $ax^2 + bx + c$ appears in the denominator of a proper rational function in reduced form. The partial fraction decomposition contains a term of the form

$$\frac{Ax + B}{ax^2 + bx + c} \quad (55)$$

where A and B are unknown coefficients to be determined.

Partial Fraction Decomposition

Let $f(x) = p(x)/q(x)$ be a proper rational function in reduced form. Assume the denominator q has been factored completely over the real numbers and m is a positive integer.

Simple linear factor A factor $x - r$ in the denominator requires the partial fraction $\frac{A}{x-r}$.

Repeated linear factor A factor $(x - r)^m$ with $m > 1$ in the denominator requires the partial fractions.

$$\frac{A_1}{(x - r)} + \frac{A_2}{(x - r)^2} + \frac{A_3}{(x - r)^3} + \cdots + \frac{A_m}{(x - r)^m} \quad (56)$$

Simple irreducible quadratic factor An irreducible factor $ax^2 + bx + c$ in the denominator requires the partial fraction

$$\frac{Ax + B}{ax^2 + bx + c} \quad (57)$$

Repeated irreducible quadratic factor An irreducible factor $(ax^2 + bx + c)^m$ with $m > 1$ in the denominator requires the partial fractions

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_mx + B_m}{(ax^2 + bx + c)^m} \quad (58)$$

8.6 Improper Integrals

Improper Integrals over Infinite Intervals

1. If f is continuous on $[a, \infty)$, then

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx \quad (59)$$

provided the limit exists.

2. If f is continuous on $(-\infty, b]$, then

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx \quad (60)$$

provided the limit exists.

3. If f is continuous on $(-\infty, \infty)$, then

$$\int_{-\infty}^\infty f(x) dx = \lim_{a \rightarrow -\infty} \int_a^c f(x) dx + \lim_{b \rightarrow \infty} \int_c^b f(x) dx \quad (61)$$

provided both limits exist, where c is any real number.

In each case, if the limit exists, the improper integral is said to **converge**, if it does not exist, the improper integral is said to **diverge**.

Improper Integrals with an Unbounded Integrand

1. Suppose f is continuous on $(a, b]$ with $\lim_{x \rightarrow a^+} f(x) = \pm \infty$. Then

$$\int_a^b f(x) dx = \lim_{c \rightarrow a^+} \int_c^b f(x) dx \quad (62)$$

provided the limit exists.

2. Suppose f is continuous on $[a, b)$ with $\lim_{x \rightarrow b^-} f(x) = \pm \infty$. Then

$$\int_a^b f(x) dx = \lim_{c \rightarrow b^-} \int_a^c f(x) dx \quad (63)$$

provided the limit exists.

3. Suppose f is continuous on $[a, b]$ except at the interior point p where f is unbounded. Then

$$\int_a^b f(x) dx = \int_a^p f(x) dx + \int_p^b f(x) dx, \quad (64)$$

provided the improper integrals on the right side exist.

In each case, if the limit exists, the improper integrals is said to **converge**, if it does not exists, the improper integral is said to **diverge**.

9 Sequences and Infinite Series

9.1 An Overview

Sequence

A **sequence** $\{a_n\}$ is an ordered list of numbers of the form

$$\{a_1, a_2, a_3, \dots, a_n, \dots\} \quad (65)$$

A sequence may be generated by a **recurrence relations** of the form $a_{n+1} = f(a_n)$, for $n = 1, 2, 3, \dots$, where a_1 is given. A sequence may also be defined with an **explicit form** of the form $a_n = f(n)$, for $n = 1, 2, 3, \dots$

Limit of a Sequence

If the terms of a sequence $\{a_n\}$ approach a unique number L as n increases, then we say $\lim_{n \rightarrow \infty} a_n = L$ exists, and the sequence **converges** to L . If the terms of the sequence do not approach a single number as n increases, the sequence has no limits, and the sequence **diverges**.

Infinite Series

Given a set of numbers $\{a_1, a_2, a_3, \dots\}$, the sum

$$a_1 + a_2 + a_3 + \dots = \sum_{k=1}^{\infty} a_k \quad (66)$$

is called an **infinity series**. Its **sequence of partial sums** $\{S_n\}$ has the terms

$$S_1 = a_1 \quad (67)$$

$$S_2 = a_1 + a_2 \quad (68)$$

$$S_3 = a_1 + a_2 + a_3 \quad (69)$$

$$\vdots \quad (70)$$

$$S_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k, \quad n = 1, 2, 3, \dots \quad (71)$$

If the sequence of partial sums $\{S_n\}$ has a limit L , the infinite series **converges** to that limits, and we write

$$\sum_{k=1}^{\infty} a_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k = \lim_{n \rightarrow \infty} S_n = L \quad (72)$$

If the sequence of partial sums diverges, the infinite series also **diverges**.

9.2 Sequences

Limits of Sequences from Limits of Functions

Suppose f is a function such that $f(n) = a_n$ for all positive integers n . If $\lim_{n \rightarrow \infty} f(x) = L$, then the limits of the sequences $\{a_n\}$ is also L .

Properties of Limits of Sequences

Assume that the sequence $\{a_n\}$ and $\{b_n\}$ have limits A and B , respectively. Then,

1. $\lim_{n \rightarrow \infty} (a_n \pm b_n) = A \pm B$
2. $\lim_{n \rightarrow \infty} ca_n = cA$, where c is a real number
3. $\lim_{n \rightarrow \infty} a_nb_n = AB$
4. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}$, provided $B \neq 0$.

Geometric Sequences

Let r be a real number. Then,

$$\lim_{n \rightarrow \infty} r^n = \begin{cases} 0 & \Leftrightarrow |r| < 1 \\ 1 & \Leftrightarrow r = 1 \\ \text{does not exist} & \Leftrightarrow r \leq -1 \vee r > 1 \end{cases} \quad (73)$$

Squeeze Theorem for Sequences

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences with $a_n \leq b_n \leq c_n$ for all n greater than some index N . If $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$.

Bounded Monotonic Sequences

A bounded monotonic sequence converges.

Growth Rates of Sequences

The following sequences are ordered according to increasing growth rates as $n \rightarrow \infty$; that is, if $\{a_n\}$ appears before $\{b_n\}$ in the list, then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\lim_{n \rightarrow \infty} \frac{b_n}{a_n} = \infty$

$$\{\ln^q n\} \ll \{n^p\} \ll \{n^p \ln^r n\} \ll \{n^{p+s}\} \ll \{b^n\} \ll \{n!\} \ll \{n^n\} \quad (74)$$

The ordering applies for $p, q, r, s, b \in \mathbb{R}^+ \wedge b > 1$.

Limit of a Sequence

The sequence $\{a_n\}$ converges to L provided the terms of a_n can be made arbitrarily close to L by taking n sufficiently large. More precisely, $\{a_n\}$ has the unique limit L if given any tolerance $\epsilon > 0$, it is possible to find a positive integer N (depending only on ϵ) such that

$$|a_n - L| < \epsilon \quad \text{whenever } n > N \quad (75)$$

if the **limit of a sequence** is L , we say the sequence **converges** to L , written

$$\lim_{n \rightarrow \infty} a_n = L \quad (76)$$

A sequence that does not converge is said to **diverge**.

9.3 Infinite Series

Geometric Series

$$\sum_{k=0}^{n-1} ar^k = S_n = a \frac{1 - r^n}{1 - r} \quad (77)$$

Geometric Series

Let $a \neq 0$ and r be real numbers. If $|r| < 1$, then $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$. If $|r| \geq 1$, then the series diverges.

9.4 The Divergence and Integral Tests

Divergence Test

If $\sum a_k$ converges, then $\lim_{n \rightarrow \infty} a_k = 0$. Equivalently, if $\lim_{n \rightarrow \infty} a_k \neq 0$, then the series diverges. However, this cannot be used to prove convergence. If $\lim_{n \rightarrow \infty} a_k = 0$, the test is inconclusive.

Harmonic Series

The harmonic series $\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots$ diverges, even though the terms of the series approach zero.

Integral Test

Suppose f is a continuous, positive, decreasing function, for $x \geq 1$, and let $a_k = f(k)$, for $k = 1, 2, 3, \dots$. Then

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \int_1^{\infty} f(x) dx \quad (78)$$

either both converge or both diverge. In the case of convergence, the value of the integral is *not*, in general, equal to the value of the series.

Convergence of the p -Series

The p -Series $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges, for $p > 1$, and diverges for $p \leq 1$.

Estimating Series with Positive Terms

Let f be continuous, positive, decreasing function, for $x \geq 1$, and let $a_k = f(k)$, for $k = 1, 2, 3, \dots$. Let $S = \sum_{k=1}^{\infty} a_k$ be a convergence series and let $S_n = \sum_{k=1}^n a_k$ be the sum of the first n terms of the series. The remainder $R_n = S - S_n$ satisfies

$$R_n \leq \int_n^{\infty} f(x) dx \quad (79)$$

Furthermore, the exact value of the series is bounded as follows:

$$S_n + \int_{n+1}^{\infty} f(x) dx \leq \sum_{k=1}^{\infty} a_k \leq S_n + \int_n^{\infty} f(x) dx. \quad (80)$$

Properties of Convergent Series

1. Suppose $\sum a_k$ converges to A and let c be a real number. The series $\sum ca_k$ converges and $\sum ca_k = c \sum a_k = cA$
2. Suppose $\sum a_k$ converges to A and $\sum b_k$ converges to B . The series $\sum(a_k \pm b_k)$ converges and $\sum(a_k \pm b_k) = \sum a_k \pm \sum b_k = A \pm B$
3. *Whether* a series converges does not depend on a finite number of terms added to or removed from the series. Specifically, if M is a positive integer, then $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=M}^{\infty} a_k$ both converge or both diverge. However, the *value* of a convergent series does change if nonzero terms are added or deleted.

9.5 The Ratio, Root, and Comparison Tests

Useful Identities

$$\lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k = \lim_{k \rightarrow \infty} \left(\frac{k+1}{k}\right)^k = e \quad (81)$$

$$\lim_{k \rightarrow \infty} k^{\frac{1}{k}} = 1 \quad (82)$$

The Ratio Test

Let $\sum a_k$ be an infinite series with positive terms and let $r = \lim_{n \rightarrow \infty} \frac{a_{k+1}}{a_k}$.

1. If $0 \leq r < 1$, the series converges.
2. If $r > 1$ (including $r = \infty$), the series diverges.
3. If $r = 1$, the test is inconclusive.

The Root Test

Let $\sum a_k$ be an infinite series with nonnegative terms and let $\rho = \lim_{n \rightarrow \infty} \sqrt[k]{a_k}$.

1. If $0 \leq \rho < 1$, the series converges.
2. If $\rho > 1$ (including $\rho = \infty$), the series diverges.
3. If $\rho = 1$, test is inconclusive.

The Comparison Test

Let $\sum a_k$ and $\sum b_k$ be a series with positive terms.

1. If $0 < a_k \leq b_k$ and $\sum b_k$ converges, then $\sum a_k$ converges.
2. If $0 < b_k \leq a_k$ and $\sum b_k$ diverges, then $\sum a_k$ diverges.

The Limit Comparison Test

Let $\sum a_k$ and $\sum b_k$ be series with positive terms and let

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = L \quad (83)$$

- If $0 < L < \infty$ (that is, L is a finite, positive number), then $\sum a_k$ and $\sum b_k$ either both converge or both diverge.
- If $L = 0$ and $\sum b_k$ converges, then $\sum a_k$ converges.
- If $L = \infty$ and $\sum b_k$ diverges, then $\sum a_k$ diverges.

Guidelines

- Begin with the Divergence Test. If you show that $\lim_{n \rightarrow \infty} a_k \neq 0$, then the series diverges and your work is finished.
- Is the series a special series? Recall the convergence properties for the following series:
 - Geometric series: $\sum ar^k$ converges for $|r| < 1$ and diverges for $|r| \geq 1$ ($a \neq 0$).
 - p -series: $\sum \frac{1}{k^p}$ converges for $p > 1$, and diverges for $p \leq 1$.
 - Check also for telescoping series.
- If the general k th term of the series looks like a function you can integrate, then try the Integral Test.
- If the general k th term of the series involves $k!$, k^k , or a^k , where a is a constant, the Ratio Test is advisable. Series with k in an exponent may yield to the Root Test.
- If the general k th term of the series is a rational function of k (or a root of a rational function), use the Comparison or the Limit Comparison Test. Use the families of series given in Step 2 as comparison series.

9.6 Alternating Series

The Alternating Series Test

The alternating series $\sum (-1)^{k+1} a_k$ converges provided

1. the terms of the series are non-increasing in magnitude ($0 < a_{k+1} \leq a_k$, for k greater than some index N) and
2. $\lim_{n \rightarrow \infty} a_n = 0$

Alternating Harmonic Series

The alternating harmonic series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$ converges (even though the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$ diverges).

Remainder in Alternating Series

Let $R_n = |S - S_n|$ be the remainder in approximating the value of a convergent alternating series $\sum_{k=1}^{\infty} (-1)^{k+1} a_k$ by the sum of its first n terms. Then $R_n \leq a_{n+1}$. In other words, the remainder is less than or equal to the magnitude of the first neglected term.

Absolute and Conditional Convergence

Assume the infinite series $\sum a_k$ converges. The series $\sum a_k$ **converges absolutely** if the series $\sum |a_k|$ converges. Otherwise, the series $\sum a_k$ **converges conditionally**.

Absolute Convergence Implies Convergence

If $\sum |a_k|$ converges, then $\sum a_k$ converges (absolute convergence implies convergence). If $\sum a_k$ diverges, then $\sum |a_k|$ diverges.

Series or Test	Form of Series	Condition for Convergence	Condition for Divergence	Comments
Geometric Series	$\sum_{k=0}^{\infty} ar^k, a \neq 0$	$ r < 1$	$ r \geq 1$	If $ r < 1$, then $\sum_{k=1}^{\infty} ar^k = \frac{a}{1-r}$
Divergence Test	$\sum_{k=1}^{\infty} a_k$	Does Not Apply	$\lim_{k \rightarrow \infty} a_k \neq 0$	Cannot be used to prove convergence.
Integral Test	$\sum_{k=1}^{\infty} a_k$, where $a_k = f(k)$ and f is continuous, positive, and decreasing.	$\int_1^{\infty} f(x)dx < \infty$	$\int_1^{\infty} f(x)dx$ does not exist	The value of the integral is not the value of the series.
p-Series	$\sum_{k=1}^{\infty} \frac{1}{k^p}$	$p > 1$	$p \leq 1$	Useful for comparison tests.
Ratio Test	$\sum_{k=1}^{\infty} a_k$, where $a_k > 0$	$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} < 1$	$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} > 1$	Inconclusive if $\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = 1$
Root Test	$\sum_{k=1}^{\infty} a_k$, where $a_k \geq 0$	$\lim_{k \rightarrow \infty} \sqrt[k]{a_k} < 1$	$\lim_{k \rightarrow \infty} \sqrt[k]{a_k} > 1$	Inconclusive if $\lim_{k \rightarrow \infty} \sqrt[k]{a_k} = 1$
Comparison Test	$\sum_{k=1}^{\infty} a_k$, where $a_k > 0$	$0 < a_k \leq b_k$ and $\sum_{k=1}^{\infty} b_k$ converges	$0 < b_k \leq a_k$ and $\sum_{k=1}^{\infty} b_k$ diverges	$\sum_{k=1}^{\infty} a_k$ is given, you supply $\sum_{k=1}^{\infty} b_k$
Limit Comparison Test	$\sum_{k=1}^{\infty} a_k$, where $a_k > 0, b_k > 0$	$0 \leq \lim_{k \rightarrow \infty} \frac{a_k}{b_k} < \infty$ and $\sum_{k=1}^{\infty} b_k$ converges	$\lim_{k \rightarrow \infty} \frac{a_k}{b_k} > 0$ and $\sum_{k=1}^{\infty} b_k$ diverges.	$\sum_{k=1}^{\infty} a_k$ is given, you supply $\sum_{k=1}^{\infty} b_k$
Alternating Series Test	$\sum_{k=1}^{\infty} (-1)^k a_k$, where $a_k > 0 < a_{k+1} \leq a_k$	$\lim_{k \rightarrow \infty} a_k = 0$	$\lim_{k \rightarrow \infty} a_k \neq 0$	Remainder R_n satisfies $R_n \leq a_{n+1}$
Absolute Convergence	$\sum_{k=1}^{\infty} a_k, a_k$ arbitrary	$\sum_{k=1}^{\infty} a_k $ converges.	Applies to arbitrary series	

10 Power Series

10.1 Approximating Functions with Polynomials

Taylor Polynomials

Let f be a function with $f', f'', \dots, f^{(n)}$ defined at a . The **n th-order Taylor polynomial** for f with its **center** at a , denoted p_n , has the property that it matches f in value, slope, and all derivatives up to the n th derivative at a ; that is,

$$p_n(a) = f(a), p'_n(a) = f'(a), \dots, p_n^{(n)}(a) = f^{(n)}(a). \quad (84)$$

The n th-order Taylor polynomial centered at a is

$$p_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x - a)^n. \quad (85)$$

More compactly, $p_n(x) = \sum_{k=0}^n c_k(x - a)^k$, where the **coefficients** are

$$c_k = \frac{f^{(k)}(a)}{k!}, \quad \text{for } k = 0, 1, 2, \dots, n \quad (86)$$

Remainder in a Taylor Polynomial

Let p_n be the Taylor polynomial of order n for f . The **remainder** in using p_n to approximate f at the point x is

$$R_n(x) = f(x) - p_n(x) \quad (87)$$

Taylor's Theorem

Let f have continuous derivatives up to $f^{(n+1)}$ on an open interval I containing a . For all x in I ,

$$f(x) = p_n(x) + R_n(x), \quad (88)$$

where p_n is the n th-order Taylor polynomial for f centered at a , and the remainder is

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{(n+1)}, \quad (89)$$

for some point c between x and a .

Estimate of the Remainder

Let n be a fixed positive integer. Suppose there exists a number M such that $|f^{(n+1)}(c)| \leq M$, for all c between a and x inclusive. The remainder in the n th-order Taylor polynomial for f centered at a satisfies

$$|R_n(x)| = |f(x) - p_n(x)| \leq M \frac{|x-a|^{n+1}}{(n+1)!} \quad (90)$$

10.2 Properties of Power Series

Power Series

A **power series** has the general form

$$\sum_{k=0}^{\infty} c_k(x-a)^k \quad (91)$$

where a and c_k are real numbers, and x is a variable. The c_k 's are the **coefficients** of the power series and a is the **center** of the power series. The set of values of x for which the series converges is its **interval of convergence**. The **radius of convergence** of the power series, denoted R , is the distance from the center of the series to the boundary of the interval of convergence.

Convergence of Power Series

A power series $\sum_{k=0}^{\infty} c_k(x-a)^k$ centered at a converges in one of three ways:

1. The series converges absolutely for all x , in which case the interval of convergence is $(-\infty, \infty)$ and the radius of convergence is $R = \infty$.
2. There is a real number $R > 0$ such that the series converges absolutely for $|x-a| < R$ and diverges for $|x-a| > R$, in which case the radius of convergence is R .
3. The series converges only at a , in which case the radius of convergence is $R = 0$.

Combining Power Series

Suppose the power series $\sum c_k x^k$ and $\sum d_k x^k$ converge absolutely to $f(x)$ and $g(x)$, respectively, on an interval I .

1. **Sum and difference:** The power series $\sum (c_k \pm d_k)x^k$ converges absolutely to $f(x) \pm g(x)$ on I .
2. **Multiplication by a power:** The power series $x^m \sum c_k x^k = \sum c_k x^{k+m}$ converges absolutely to $x^m f(x)$ on I , provided m is an integer such that $k+m \geq 0$ for all terms of the series.

3. **Composition:** If $h(x) = bx^m$, where m is a positive integer and b is a real number, the power series $\sum c_k(h(x))^k$ converges absolutely to the composite function $f(h(x))$, for all x such that $h(x)$ is in I .

Differentiating and Integrating Power Series

Let the function f be defined by the power series $\sum c_k(x - a)^k$ on its interval of convergence I .

- f is a continuous function on I .
- The power series may be differentiated or integrated term by term, and the resulting power series converges to $f'(x)$ or $\int f(x) dx + C$, respectively, at all points in the interior of I , where C is an arbitrary constant.

10.3 Taylor Series

Taylor/Maclaurin Series for a Function

Suppose the function f has derivatives of all orders on an interval containing the point a . The **Taylor series for f centered at a** is

$$f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f^{(3)}(a)}{3!}(x - a)^3 + \cdots \quad (92)$$

$$= \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x - a)^k. \quad (93)$$

A Taylor series centered at 0 is called a **Maclaurin series**.

Binomial Coefficients

$$\forall p, k \in \mathbb{R} \wedge k \geq 1$$

$$\binom{p}{k} = \frac{p(p-1)(p-2) \cdots (p-k+1)}{k!}. \quad (94)$$

With the special case of $\binom{p}{0} = 1$.

Binomial Series

$\forall p \in \mathbb{R} \wedge p \neq 0$, the Taylor series for $f(x) = (1+x)^p$ centered at 0 is the **binomial series**

$$\sum_{k=0}^{\infty} x^k = \sum_{k=0}^{\infty} \frac{p(p-1)(p-2) \cdots (p-k+1)}{k!} x^k \quad (95)$$

$$= 1 + px + \frac{p(p-1)(p-2)}{3!} x^3 + \cdots \quad (96)$$

The series converges for $|x| < 1$ (and possibly at the endpoints, depending on p). If p is a nonnegative integer, the series terminates and results in a polynomial of degree p .

Convergence of Taylor Series

Let f have derivatives of all orders on an open interval I containing a . The Taylor series for f centered at a converges to f , for all x in I , if and only if $\lim_{n \rightarrow \infty} R_n(x) = 0$, for all x in I , where

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \quad (97)$$

is the remainder at x (with c between x and a).

Taylor Series Functions

11 Parametric and Polar Curves

11.1 Parametric Equations

Forward or Positive Orientation

The direction in which a parametric curve is generated as the parameter increases is called the **forward**, or **positive, orientation** of the curve.

Derivative for Parametric Curves

Let $x = g(t)$ and $y = h(t)$, where g and h are differentiable on an interval $[a, b]$. Then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{y'(t)}{x'(t)} \quad (98)$$

provided $\frac{dx}{dt} \neq 0$.

11.2 Polar Coordinates

Introduction

- Up to now we have only studied in a Cartesian coordinate system.
 - A Cartesian coordinate system is just a plane described by Cartesian (or, algebraic) equations and points in a finite dimensions.
 - * *One Dimension*: Lines.
 - * *Two Dimensions*: x^2 .
 - * *Three, Four*: Upper-level Calculus and Physics.
- Let's define an alternative coordinate system — **polar coordinate**.
 - coordinates are constants on circles and rays.
 - Useful for navigation, position, and gravitation fields.

Defining Polar Coordinates

Pole The origin of the coordinate system.

Polar Axis Synonymous for the positive x -axis.

Polar Coordinates A polar coordinates P has the form (r, θ) .

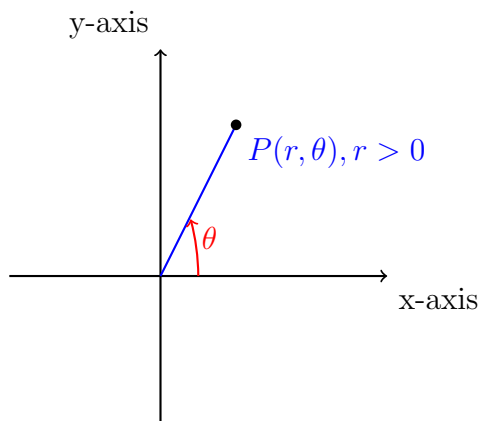
Radial Coordinate The radial coordinate r describes the *signed*, or *directed*, distance from the origin to P .

Angular Coordinate The angular coordinate θ describes an angle whose initial side is the positive x -axis and whose terminal side lies on the ray passing through the origin and P .

Notes

- **Positive angle measurements are measured *counterclockwise* from the origin.**
- Every point has multiple representations.
 - Angles are periodic, so multiples of 2π gives the same angle.

- Coordinates may be negative. So (r, θ) can be represented as $(-r, \theta + \pi)$ and $(-r, \theta - \pi)$



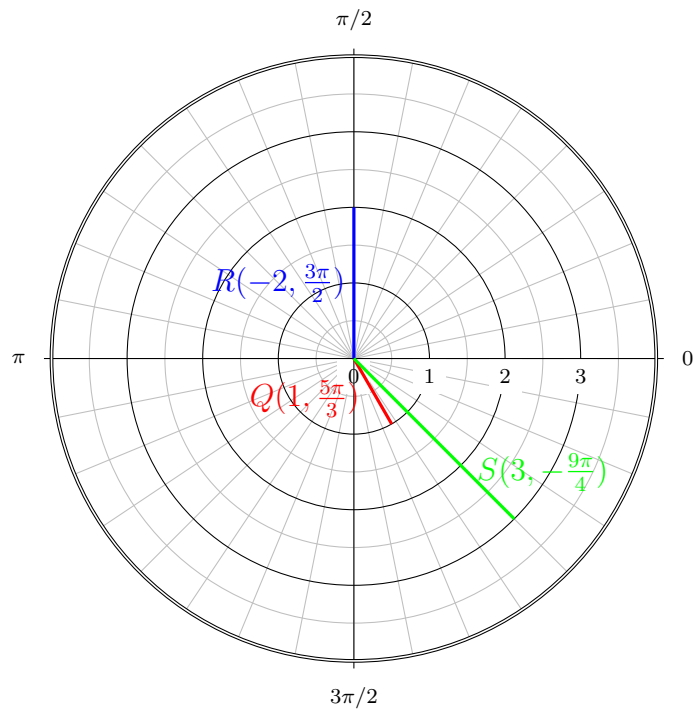
In summary,

- $(r, \theta + 2\pi)$ represents the same point as (r, θ)
- $P(r, \theta)$ and $P'(-r, \theta)$ are reflections through the origin.

Examples

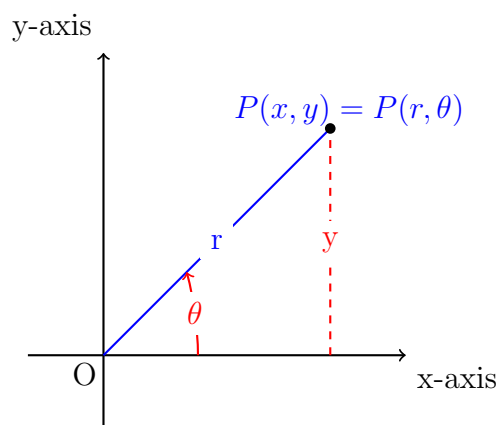
Graph the following points in polar coordinates:

- $Q(1, \frac{5\pi}{3})$
- $R(-2, \frac{3\pi}{2})$
- $S(3, -\frac{9\pi}{4})$
 - Now give two alternative representations.
 - $S'(3, \frac{1\pi}{4})$
 - $S''(-3, -\frac{5\pi}{4})$



Converting Between Cartesian and Polar Coordinates

- We sometimes need to convert between Cartesian and polar coordinates.
- Let's turn this problem into a right triangle.



A point with polar coordinates (r, θ) has Cartesian coordinates (x, y) , where

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta \quad (99)$$

A point with Cartesian coordinates (x, y) has polar coordinates (r, θ) , where

$$r^2 = x^2 + y^2 \quad \text{and} \quad \tan \theta = \frac{y}{x} \quad (100)$$

Examples

BE SURE TO GRAPH POINTS IN CARTESIAN FIRST.

Polar to Cartesian

Express the point with the following polar coordinates in Cartesian coordinates: $P(3, \frac{2\pi}{3})$

$$\begin{aligned} x &= r \cos \theta & y &= r \sin \theta \\ &= 3 \cos(2\pi/3) & &= 3 \sin(2\pi/3) \\ &= -3(1/2) & &= 3(\sqrt{3}/2) \\ &= -3/2 & &= 3\sqrt{3}/2 \end{aligned}$$

Polar to Cartesian

Express the point with the following polar coordinates in Cartesian coordinates: $Q(e, -\frac{\pi}{4})$

$$\begin{aligned} x &= r \cos \theta & y &= r \sin \theta \\ &= e \cos(-\pi/4) & &= e \sin(-\pi/4) \\ &= e(\sqrt{2}/2) & &= -e(\sqrt{2}/2) \\ &= e\sqrt{2}/2 & &= -e\sqrt{2}/2 \end{aligned}$$

Cartesian To Polar

Express the point with the following Polar coordinates to Cartesian coordinates: $R(1, -1)$

$$\begin{array}{ll}
r &= \sqrt{x^2 + y^2} \\
&= \sqrt{1^2 + (-1)^2} \\
&= \sqrt{2} \\
\tan \theta &= y/x \\
&= -1/1 \\
&= -1 \\
\theta &= -\pi/4 \text{ or } 7\pi/4.
\end{array}$$

Therefore, two possible solutions are: $(\sqrt{2}, -\pi/4)$ or $(\sqrt{2}, 7\pi/4)$

Cartesian To Polar

Express the point with the following Polar coordinates to Cartesian coordinates: $S(1, \sqrt{3})$

$$\begin{array}{ll}
r &= \sqrt{x^2 + y^2} \\
&= \sqrt{(\sqrt{3})^2 + (1)^2} \\
&= 2 \\
\tan \theta &= y/x \\
&= \sqrt{3}/1 \\
\theta &= \pi/3 \text{ or } 4\pi/3.
\end{array}$$

Therefore, two possible solutions are: $(2, \pi/3)$ or $(2, 4\pi/3)$.

Basic Curves in Polar Coordinates

- A curve in polar coordinates is the set of **points** that satisfy an equation in r and θ .
- This makes graphing some things easier than others.
- Look at $r = 3$ is the set of all points that satisfy being away from the origin of 3 units.
 - This is because θ is not specified, it's arbitrary. Basically, θ is the function.
 - In general, $r = a, \forall a \in \mathbb{R}^+$ describes a circle.
- Taking the converse, let r be arbitrary.
 - If the r is arbitrary, and we specify the angle, what do you think we get?

- A line!
- Take $\sqrt{2}/2$.

Polar to Cartesian Graph Example

Convert the polar equation $r = 6 \sin \theta$ to Cartesian coordinates and describe the corresponding graph.

$$r^2 = 6r \sin \theta \quad (101)$$

$$x^2 + y^2 = 6y \quad (102)$$

$$0 = x^2 + y^2 - 6y \quad (103)$$

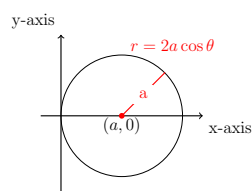
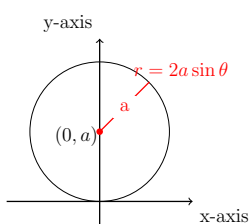
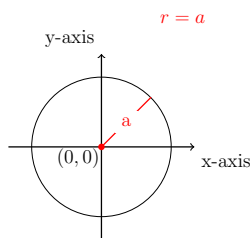
$$= x^2 + (y^2 - 6y + 9) - 9 \quad (104)$$

$$= x^2 + (y - 3)^2 - 9 \quad (105)$$

We recognize this to be the equation of a circle, centered at $(0, 3)$ at 3. We can also generalize this.

Circle in Polar Coordinates

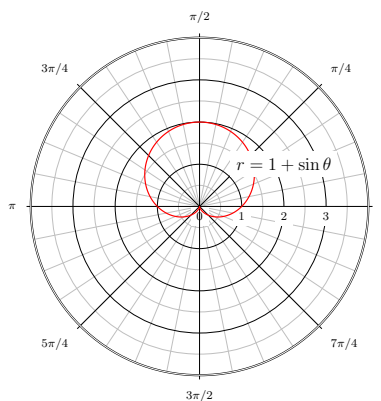
- The equation $r = a$ describes a circle of radius $|a|$ centered at $(0, 0)$.
- The equation $r = 2a \sin \theta$ describes a circle of radius $|a|$ centered at $(0, a)$.
- The equation $r = 2a \cos \theta$ describes a circle of radius $|a|$ centered at $(a, 0)$.



Graphing In Polar Coordinates

Graph the polar equation $r = f(\theta) = 1 + \sin \theta$

θ	$r = 1 + \sin\theta$
0	1
$\pi/6$	$3/2$
$\pi/2$	2
$5\pi/6$	$3/2$
π	1
$7\pi/6$	$1/2$
$3\pi/2$	0
$11\pi/6$	$1/2$
2π	1



The resulting curve is known as a **cardioid**.

Cartesian-to-Polar Method for Graphing $r = f(\theta)$

1. Graph $r = f(\theta)$ as if r and θ were Cartesian coordinates with θ on the horizontal axis and r on the vertical axis. Be sure to choose an interval in θ on which the entire polar curve is produced.
2. Use the Cartesian graph in Step 1 as a guide to sketch the points (r, θ) on the final *polar* curve.

Example

With the alternate graphing method, graph $r = 1 + \sin\theta$.

Symmetry In Polar Equations

Symmetry about the x-axis occurs if the point (r, θ) is on the graph whenever $(r, -\theta)$ is on the graph.

Symmetry about the y-axis occurs if the point (r, θ) is on the graph whenever $(r, \pi - \theta) = (-r, -\theta)$ is on the graph.

Symmetry about the origin occurs if the point (r, θ) is on the graph whenever $(-r, \theta) = (r, \theta + \pi)$ is on the graph.

11.3 Calculus in Polar Coordinates

Slope of a Tangent Line

Let f be a differentiable function at θ_0 . The slope of the line tangent to the curve $r = f(\theta)$ at the point $(f(\theta_0), \theta_0)$ is

$$\frac{dy}{dx} = \frac{f'(\theta_0) \sin \theta_0 + f(\theta_0) \cos \theta_0}{f'(\theta_0) \cos \theta_0 - f(\theta_0)} \quad (106)$$

Area of Regions in Polar Coordinates

Let R be the region bounded by the graphs of $r = f(\theta)$ and $r = g(\theta)$, between $\theta = \alpha$ and $\theta = \beta$, where f and g are continuous and $f(\theta) \geq g(\theta) \geq 0$ on $[\alpha, \beta]$. The area of R is

$$\int_{\alpha}^{\beta} \frac{1}{2} (f(\theta)^2 - g(\theta)^2) d\theta \quad (107)$$